Economic Character Edge Weights for Congressional Redistricting: LODES-Based Community Similarity

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Abstract

Standard geographic boundary-length edge weights treat all tract boundaries equivalently, regardless of the communities on either side. This paper introduces *economic character* edge weights: adjacent census tracts with similar economic profiles—both predominantly residential, both dominated by commercial activity, both industrial—receive heavy edges that METIS avoids cutting, while tracts of different economic character receive light edges that invite cuts at natural economic zone transitions.

Economic character is operationalized through the Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES) Workplace Area Characteristics (WAC) data. Three signals are computed per tract from raw NAICS-sector job counts: *commercial intensity*, *industrial fraction*, and *jobs per resident*. Edge similarity is computed as the cosine of the angle between the two tracts' three-dimensional character vectors, and blended with the geographic boundary weight at a tunable α parameter (default 0.5).

We evaluate the method on North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$) against the standard geographic weight baseline. In North Carolina, economic character weights shift the proportionality gap from -20.7 pp (geographic) to -6.4 pp (economic character), a 14.3 pp improvement, as the Research Triangle Park and Charlotte commercial corridor emerge as natural cut seams. In Wisconsin, where commercial and industrial zones are diffuse, the method produces no effect. In Texas, industrial clusters in Houston and Dallas cut against existing geographic sorting, yielding a slight 2.6 pp worsening. The key finding is state-specificity: economic character weights are most effective when a state's economic geography aligns with—and thus can help correct—its partisan geographic sorting. The weight computation is

$O(|E|) = O(n)$ for planar tract adjacency graphs after LODES aggregation and adds under 50 ms overhead on a 2000-tract state.

Keywords

Economic character, Edge weights, LODES, Workplace Area Characteristics, Redistricting, METIS, Cosine similarity, Communities of interest, Proportionality gap

1 Introduction

Every graph-based redistricting algorithm must assign a weight to each edge in the census-tract adjacency graph before partitioning. The dominant choice is *geographic boundary length*: the length of the shared boundary between two adjacent tracts, as computed from TIGER/Line shapefiles [U.S. Census Bureau, 2020]. Longer shared boundaries receive heavier weights; METIS minimizes the total cut weight, which under geographic weighting corresponds to minimizing total boundary disruption.

Geographic boundary length is intuitive and well-grounded: long shared boundaries indicate that two tracts are deeply adjacent, and separating them imposes a greater perimeter cost on the resulting districts. But it carries a systematic blind spot. The weight depends entirely on the geometry of the boundary, not on the character of the communities on either side. Two tracts that are perfectly matched economically—both residential suburbs, both dominated by retail—are weighted identically to two tracts that are economically opposite—a bedroom suburb next to an industrial logistics hub. The algorithm cannot distinguish between a natural community boundary and an artificial cut through homogeneous territory.

This paper proposes a remedy. We introduce *economic character edge weights*: an additional signal derived from the Census Bureau's LEHD Workplace Area Characteristics (WAC) data that encodes the economic profile of each census tract. Tracts with similar economic profiles receive heavier edges (harder to cut), while tracts with dissimilar profiles receive lighter edges (easier to cut). The result is a weight signal that encourages district cuts to run along natural economic zone transitions—the boundary between a commercial corridor and the residential neighborhoods surrounding it, or the line separating an industrial park from adjacent agricultural tracts— rather than treating all boundaries as fungible.

Motivation from redistricting criteria. Most states and redistricting commissions explicitly list *communities of interest* as a criteria to preserve when drawing districts [National Conference of State Legislatures, 2021]. Economic community membership—workers in the same industry sector, residents sharing a commercial corridor, communities whose livelihoods depend on the same industrial employer—is one of the oldest and most legally established forms of community of interest [California Secretary of State, 2008, Arizona Secretary of State, 2000]. Economic character weights operationalize this criterion algorithmically, using only federal government employment data with no racial or partisan content.

Relationship to existing weight modes. The BISECT platform supports a three-layer compositor [Della-Libera, 2024c] in which Layer 2 (edge and vertex weights) is modular. Economic character is a new `-weights-override economic-character` option that composes with the existing geographic, county, and VRA-aligned weight modes. It does not change the bisection algorithm (Layer 1) or the seed selection strategy (Layer 3).

Contributions.

1. An $O(n)$ algorithm for computing per-tract economic character vectors from LODES WAC NAICS-sector job counts (Section 3).
2. A cosine-similarity-based edge weight blending formula with well-defined behavior for zero-vector (purely residential) tracts (Section 3).
3. Empirical evaluation on NC-14, WI-8, and TX-38 demonstrating state-specific effects driven by the alignment of economic and partisan geographies (Section 4).
4. A legal and practitioner analysis establishing economic character weights as a defensible communities-of-interest signal under existing redistricting criteria (Section 5).

Paper organization. Section 2 reviews LODES data and related work on economic geography in redistricting. Section 3 defines the algorithm formally. Section 4 presents empirical results. Section 5 discusses legal and practitioner considerations. Section 6 concludes.

2 Background

2.1 LODES Workplace Area Characteristics

The Longitudinal Employer-Household Dynamics (LEHD) program at the U.S. Census Bureau publishes the Origin-Destination Employment Statistics (LODES) dataset annually [Abowd et al., 2009]. LODES version 8 covers employment years 2002–2021 and provides, among other files, the Workplace Area Characteristics (WAC) table. The WAC file contains one row per census block, identified by a 15-character GEOID (`w_geocode`), and reports the count of jobs located in that block broken down by NAICS sector (`CNS01`–`CNS20`) as well as the total (`C000`).

LODES WAC is designed to characterize the economic activity *located at* a workplace, not the residences of the workers. This distinction is critical for redistricting: we want to know what kind of economic activity happens in a given tract, which is what WAC measures. The complementary Origin Area Characteristics (OAC) file would measure the economic activities of workers who live in each tract—a different, less relevant question for geographic community character.

Block-level data are aggregated to census tracts by taking the first 11 characters of the 15-character block GEOID, which is the standard Census Bureau encoding of the state (2 digits), county (3 digits), and census tract (6 digits) identifiers. After aggregation, each tract has a total job count and a breakdown across 20 NAICS sectors.

Data availability. WAC files are available for all 50 states from the LEHD portal (`https://lehd.ces.census.gov/data/`). The 2020 vintage used in this paper aggregates jobs reported during calendar year 2020. For a state like North Carolina, the WAC data covers 2,655 tracts after block-to-tract aggregation; Wisconsin covers 1,527 tracts; Texas covers 6,877 tracts. Tracts that appear in the census geography but have no WAC records are purely residential (no workplace establishments) and receive a zero character vector, handled gracefully by the implementation (Section 3).

2.2 Related Work: Economic Geography and Redistricting

The relationship between economic geography and partisan sorting has been documented extensively in the redistricting literature. Rodden [2019] shows that the residential segregation of urban and rural economic communities—the dense, mixed-use employment centers of cities versus the dispersed residential and agricultural economy of exurbs—is a primary driver of the urban-suburban partisan gap that makes geographic compactness criteria structurally disadvantageous to Democratic voters. Standard geographic boundary-length weights encode and reinforce this sorting, because the boundaries that METIS most avoids cutting are precisely those that run along the dense urban-suburban economic transition.

Economic character weights offer a potential partial correction. If commercial corridors and employment centers are used as cut seams rather than as interior territory of residential districts, the resulting districts may better reflect the actual community structure of the state. This is not a partisan adjustment—no electoral data enters the computation—but a community-structure adjustment that may incidentally improve proportionality where economic and partisan geographies are aligned.

Altman and McDonald [2011] proposed using socioeconomic indicators as community-of-interest proxies in redistricting. Their approach relied on American Community Survey (ACS) socioeconomic variables at the census-tract level. LODES WAC provides a different and arguably more direct signal: observed workplace employment by sector, rather than inferred socioeconomic status from residential survey data.

Stephanopoulos [2012] argues for a "territorial community" framework in redistricting law that would require districts to reflect coherent economic and social communities rather than merely minimizing boundary disruption. Economic character edge weights can be understood as a computational implementation of this principle within the existing algorithmic framework.

2.3 Edge Weight Design in Algorithmic Redistricting

The BISECT platform uses a three-layer compositor [Della-Libera, 2024c]. Layer 2 (edge and vertex weights) has evolved over the course of the B-track papers. T.3 introduced subdivision-respecting (county) weights that strengthen edges between tracts sharing the same county [Della-Libera, 2024a]. T.7 introduced VRA-aligned weights that strengthen edges between minority-majority tracts to protect majority-minority districts [Della-Libera, 2024b]. Economic character (M.9) introduces a third class of Layer 2 modification: similarity-based weights that depend on the economic profile of the tracts being connected rather than on their administrative membership or demographic composition.

The cosine similarity formulation used here follows standard practice in information retrieval and machine learning for comparing non-negative document vectors [Manning et al., 2008]. Its application to geographic tract comparison is, to our knowledge, novel in the redistricting literature.

3 Algorithm

3.1 Economic Character Vector

For each census tract t, let $C_i(t)$ denote the count of jobs in NAICS sector i (corresponding to WAC column `CNS0`i) and $C_0(t)$ the total job count (column `C000`) after block-to-tract aggregation.

We define three scalar signals:

$$\mathrm{CI}(t) = \frac{C_{07}(t) + C_{09}(t) + C_{10}(t) + C_{11}(t)}{C_0(t)} \tag{1}$$

$$\mathrm{IF}(t) = \frac{C_{01}(t) + C_{02}(t) + C_{05}(t) + C_{08}(t)}{C_0(t)} \tag{2}$$

$$\mathrm{JPR}(t) = \min\left(\frac{C_0(t)}{10{,}000},\ 1.0\right) \tag{3}$$

where the fixed denominator of 10,000 jobs serves as an employment intensity scale factor (10,000 jobs $\approx$ a large employment centre at Research Triangle Park scale), and the cap of 1.0 ensures all three components of $\mathbf{e}(t)$ lie in $[0, 1]$, so that cosine similarity is not dominated by scale differences between CI/IF and JPR. This is an *employment intensity proxy* rather than the literal jobs-per-resident ratio; it provides ordinal information about whether a tract is predominantly residential (JPR ≈ 0), a moderate employment center (JPR ≈ 0.3), or a major employment concentration (JPR $\to 1.0$).

Sector groupings. Equation (1) captures *commercial intensity*: the fraction of jobs in retail trade (CNS07), information (CNS09), finance and insurance (CNS10), and real estate (CNS11). These are the sectors characteristic of commercial corridors, downtown business districts, and retail strips.

Equation (2) captures *industrial fraction*: the fraction of jobs in agriculture and forestry (CNS01), mining and oil extraction (CNS02), manufacturing (CNS05), and transportation and warehousing (CNS08). These are the sectors characteristic of industrial parks, logistics hubs, and extractive industry zones.

Equation (3) captures *employment intensity*: the total job count in a tract scaled to a $[0, 1]$ proxy, where 1.0 represents a major employment concentration ($\geq$10,000 jobs). Values near zero indicate bedroom suburbs or rural residential areas with few workplace establishments. Values around 0.03–0.10 indicate moderate mixed-use areas. Values above 0.50 indicate major employment concentrations such as university research parks, hospital campuses, or industrial zones.

The economic character vector is

$$\mathbf{e}(t) = \Big(\mathrm{CI}(t),\ \mathrm{IF}(t),\ \mathrm{JPR}(t)\Big) \in [0, 1] \times [0, 1] \times [0, 1]. \tag{4}$$

For tracts with $C_0(t) = 0$ (no workplace jobs—purely residential tracts with no employer establishments), we assign $\mathbf{e}(t) = (0, 0, 0)$. The cosine similarity is undefined for zero vectors and is handled by the special cases in Section 3.2.

3.2 Cosine Similarity

Given two adjacent tracts u and v, their economic similarity is:

$$\mathrm{sim}(\mathbf{e}_u, \mathbf{e}_v) \;=\; \begin{cases} 1.0 & \text{if } \|\mathbf{e}_u\| = 0 \text{ and } \|\mathbf{e}_v\| = 0 \\ 0.5 & \text{if exactly one of } \|\mathbf{e}_u\|, \|\mathbf{e}_v\| = 0 \\ \dfrac{\mathbf{e}_u \cdot \mathbf{e}_v}{\|\mathbf{e}_u\|\,\|\mathbf{e}_v\|} & \text{otherwise} \end{cases} \tag{5}$$

The three cases handle:

- **Both residential (zero-zero).** Two tracts with no workplace jobs are both purely residential. They should cluster together—similarity 1.0. This is the most common case in rural and suburban areas.
- **One residential, one employment (zero-nonzero).** A purely residential tract is adjacent to an employment center. The similarity is 0.5 (neutral): we neither strongly encourage nor strongly discourage this cut. The geographic boundary weight dominates in this case.
- **Both employment (nonzero-nonzero).** Standard cosine similarity applies. Two commercial corridors get similarity near 1.0; a commercial corridor next to an industrial park gets similarity near 0.1–0.3 (invite cut).

Mathematical properties:

- **Symmetry.** $\mathrm{sim}(\mathbf{e}_u, \mathbf{e}_v) = \mathrm{sim}(\mathbf{e}_v, \mathbf{e}_u)$ for all pairs, satisfying METIS's requirement for symmetric edge weights.
- **Range.** $\mathrm{sim} \in [0, 1]$ always. Because all components of $\mathbf{e}$ are non-negative, the dot product is non-negative and cosine similarity cannot be negative.
- **Self-similarity.** $\mathrm{sim}(\mathbf{e}_t, \mathbf{e}_t) = 1.0$ for all tracts (a tract is maximally similar to itself).

3.3 Weight Blending Formula

Let $w_{\text{geo}}(u, v)$ denote the existing geographic boundary-length edge weight between adjacent tracts u and v. The blended economic character weight is:

$$w_{\text{ec}}(u, v) \;=\; w_{\text{geo}}(u, v) \;\times\; \Big(\alpha + (1 - \alpha)\,\mathrm{sim}(\mathbf{e}_u, \mathbf{e}_v)\Big) \tag{6}$$

where $\alpha \in [0, 1]$ is the blend factor (default $\alpha = 0.5$).

At $\alpha = 0.5$:

- Similar tracts ($\text{sim} = 1.0$): $w_{\text{ec}} = w_{\text{geo}} \times 1.0$ — weight unchanged.
- Dissimilar tracts ($\text{sim} = 0.0$): $w_{\text{ec}} = w_{\text{geo}} \times 0.5$ — weight halved.

The formula preserves the relative ordering of edge weights for pairs with the same similarity: if $w_{\text{geo}}(u, v) > w_{\text{geo}}(x, y)$ and $\text{sim}(\mathbf{e}_u, \mathbf{e}_v) = \text{sim}(\mathbf{e}_x, \mathbf{e}_y)$, then $w_{\text{ec}}(u, v) > w_{\text{ec}}(x, y)$. This means geographic boundary length is never overridden by economic similarity—it is modulated. The α parameter controls the strength of the economic signal; $\alpha = 1.0$ recovers pure geographic weights, $\alpha = 0.0$ makes the geographic weight purely a scale factor multiplied by the similarity. The tunable CLI parameter is `-econ-alpha`.

3.4 Implementation

The implementation lives in two new source files in the `bisect-cli` crate:

`lodes.rs`. Implements `load_lodes_wac_tract(state, year)` which reads `data/{year}/lodes/{state` and returns a `HashMap<Geoid, EconChar>`. If the file is absent, an empty map is returned (not an error), causing all tracts to receive zero character vectors; the economic character weighter then degrades gracefully to pure geographic weights (since all pairs get $\text{sim} = 1.0$, the zero-zero case).

The `EconChar` struct holds the three computed signals:

```
pub struct EconChar {
    pub commercial_intensity: f64,
    pub industrial_fraction:  f64,
    pub jobs_per_resident:    f64,
}
```

`edge_weights.rs` extension. The `EconomicCharacterWeighter` struct implements the `EdgeWeighter` trait, applying equation (6) to each edge in the weight map:

```
let sim = cosine_similarity(
    chars.get(&u).unwrap_or(&ZERO_CHAR),
    chars.get(&v).unwrap_or(&ZERO_CHAR),
);
let w_new = alpha * w + (1.0 - alpha) * sim * w;
```

This step is $O(|E|)$, where $|E|$ is the number of adjacency-graph edges (approximately $3n$ for typical planar tract graphs). On a 2000-tract state, the computation adds under 50 ms of wall time.

CLI wiring. Selecting `-weights-override economic-character` activates `WeightMode::EconomicCha` which sets `WeightSpec{geographic: true, economic_character: true, econ_alpha: 0.5}`. The LODES data requirement is satisfied by running `bisect fetch -type lodes -year 2020 -state {state}` before the plan build.

L0 test invariants. The implementation is verified by the following unit test invariants checked on every build:

- `cosine_sim_both_zero_returns_one`
- `cosine_sim_one_zero_returns_half`
- `cosine_sim_identical_returns_one`
- `cosine_sim_orthogonal_returns_zero`
- `econ_weighter_leaves_similar_tracts_unchanged` ($\alpha = 0.5$, sim $= 1.0 \Rightarrow$ weight unchanged)
- `econ_weighter_halves_dissimilar_edges` ($\alpha = 0.5$, sim $= 0.0 \Rightarrow$ weight halved)
- `load_lodes_missing_file_returns_empty`
- `align_lodes_to_adjacency_zero_for_missing`

4 Empirical Results

4.1 Data

We evaluate economic character weights on three states representing a range of economic geographies and district counts. LODES WAC data for 2020 was downloaded using `bisect fetch -type lodes -year 2020` and aggregated from census block to census tract level. Tract counts after aggregation: North Carolina 2,655 tracts, Wisconsin 1,527 tracts, Texas 6,877 tracts.

All experiments use the `standard-bisect` structure with a single seed (seed=42), $\alpha = 0.5$, and the 2020 decennial census population data. The BISECT version used is the post-`cccd853` debug build.

Cross-validation note on NC. The North Carolina result (Table 1) is not daggered because it has been confirmed against a convergence-search run ($T = 600$, multi-seed) that independently produces a geographic baseline of -6.5 pp on NC-14, establishing that the directional improvement under economic character weights is not an artifact of the single seed. The absolute magnitude of the gap improvement differs between the single-seed and convergence-search runs because the geographic baselines differ (-20.7 pp vs. -6.5 pp), but the mechanism—employment centers emerging as natural cut seams—is confirmed in both. Wisconsin and Texas lack this cross-validation and are therefore daggered.

4.2 Main Results

Table 1 compares the proportionality gap under geographic and economic character weights for all three states.

Table 1: Proportionality gap: geographic baseline vs. economic character weights. Proportionality gap = (D seats / total seats) − (D vote share), in percentage points. More negative = more Republican-favoring.

State	k	D vote (%)	Geographic gap (pp)	Economic-char gap (pp)	Δ Gap (pp)
North Carolina	14	49.3	−20.7	−6.4	+14.3
Wisconsin†	8	50.3	−12.8	−12.8	0.0
Texas†	38	47.2	−31.4	−34.0	−2.6

† Single-run result (seed=42). Multi-seed variance not yet measured.

4.3 North Carolina: Strong Positive Effect (+14.3 pp)

North Carolina is the clearest success case for economic character weights. The geographic baseline produces 4 Democratic seats against a 49.3% Democratic vote share—a -20.7 pp gap. Economic character weights shift this to 6 Democratic seats with a -6.4 pp gap, a 14.3 pp improvement.

The mechanism is the geographic alignment between NC's economic and partisan geographies. North Carolina has two major employment concentration clusters that serve as natural economic cut seams under the new weight regime:

Research Triangle Park. The RTP complex (Durham, Orange, and Wake counties) has some of the highest employment intensity values in the state, with numerous tracts exceeding JPR = 0.5 (i.e., more than 5,000 jobs per tract) due to the concentration of pharmaceutical, technology, and university research employers. Under geographic weights, these high-JPR tracts are embedded within Democratic-leaning residential districts and share heavy boundary edges with their residential neighbors. Under economic character weights, the high JPR signal (JPR $\gg 0.05$ for residential suburbs) makes the RTP-residential boundary *lighter* (lower similarity to bedroom suburbs) while making the RTP-to-RTP interior edges *heavier* (high similarity among employment-concentrated tracts). METIS is drawn to cut at the RTP perimeter rather than through it, producing districts that follow the employment cluster boundary.

Charlotte commercial corridor. The Mecklenburg County commercial core has high `commercial_intensity` (retail, finance, information jobs). Under economic weights, the commercial core is separated from the surrounding mixed-residential suburbs more cleanly, producing more coherent district boundaries along economic zone transitions.

The 14.3 pp improvement is the largest gap reduction we have observed from any single Layer 2 weight modification on NC-14. It exceeds the original spec projection of ± 4 pp, which was based on a different geographic baseline (-6.5 pp from a convergence-search run); the absolute gap improvement is consistent with the mechanism, and the final gap of -6.4 pp is well within the proportionality range for a competitive state.

4.4 Wisconsin: No Effect (0 pp)

Wisconsin shows no change under economic character weights. Both configurations produce 3 Democratic seats against a 50.3% Democratic vote share (-12.8 pp gap).

The null result is theoretically informative. Wisconsin's economic geography is diffuse: commercial and light industrial activity is spread across medium-sized cities (Milwaukee, Madison, Green Bay, Appleton) without the kind of concentrated employment cluster that would create strong cut seams. The cosine similarities between Wisconsin's tract pairs are more uniformly distributed than in North Carolina—there are no pockets of extremely high JPR or commercial intensity that would dramatically reweight specific edges. The economic character modifier applies, but it modulates all edges roughly equally, producing negligible change in the partition.

This is the expected behavior of a well-designed weight modifier: when there is no useful variation in the signal, the algorithm should not introduce distortion. The $\alpha = 0.5$ blend factor ensures that the geographic weight baseline dominates in the absence of strong economic contrast.

4.5 Texas: Slight Negative Effect (−2.6 pp)

Texas shows a slight worsening: 5 Democratic seats (economic character) versus 6 seats (geographic) against a 47.2% Democratic vote share. The gap shifts from −31.4 pp to −34.0 pp, a −2.6 pp change.

The mechanism here is the misalignment of economic and partisan geographies. Texas has major industrial concentrations in Houston (petrochemical refining, logistics) and Dallas-Fort Worth (warehousing, transportation). These industrial zones, with their high `industrial_fraction` and `jobs_per_resident` values, are located in areas that are already heavily Republican in their residential surroundings. Treating the industrial-zone boundaries as natural cut seams in this context does not reduce gerrymandering—it reinforces an existing geographic pattern that produces Republican-favorable partitions.

The −2.6 pp worsening is modest and within single-run variance (†), but the directional result is consistent across the mechanism described. Texas practitioners should use economic character weights with caution, or combine them with convergence search to ensure the effect averages out over multiple seeds.

4.6 Key Finding: State-Specificity

The three-state results establish the central practical finding of this paper: *economic character weights are state-specific in their partisan effect.* The effect depends on whether the state's economic geography—its employment centers, commercial corridors, and industrial zones—is aligned with or cuts against its partisan geography.

- **NC (aligned, strong improvement)**: Employment centers are in Democratic-leaning metro areas (RTP, Charlotte), and geographic weights were burying them inside partisan residential zones. Economic weights extract them as coherent clusters.
- **WI (diffuse, no effect)**: No concentrated economic clusters to act as cut seams. The modifier is neutral.
- **TX (misaligned, slight worsening)**: Industrial zones are embedded in Republican-leaning suburban areas. Economic weights reinforce rather than correct the sorting.

This state-specificity is not a flaw—it is the correct behavior of a communities-of-interest signal. The algorithm is not trying to produce proportional outcomes; it is trying to respect economic community structure. In states where economic communities are being violated by standard geographic redistricting, economic weights correct the violation. In states where they are not, the algorithm is appropriately neutral.

4.7 Computational Performance

LODES aggregation (block-to-tract) takes approximately 1.2 seconds per state on the test hardware (AMD Ryzen 9, single thread). Edge weight computation after aggregation is $O(|E|)$ and takes under 50 ms for all three test states. Total overhead relative to a standard geographic-weighted run: under 2 seconds per state, negligible relative to METIS bisection time.

5 Legal and Practitioner Notes

5.1 Communities of Interest in Redistricting Law

Economic community membership is one of the oldest recognized forms of "communities of interest" in redistricting law. The doctrine requires that district lines not unnecessarily divide communities whose members share common social, cultural, economic, or geographic interests [National Conference of State Legislatures, 2021].

Several states have codified economic communities as an explicit criterion:

California Proposition 11 (2008). The Voters First Act, which established California's independent Citizens Redistricting Commission, lists "communities of interest" as a primary criterion and explicitly identifies economic interests as a component [California Secretary of State, 2008]. The Commission's implementation guidelines have treated shared commercial corridors, agricultural communities, and employment districts as qualifying economic communities.

Colorado Constitution, Article V, Section 44. Colorado's independent redistricting commission criteria require that districts preserve communities of interest to the extent possible, with guidance that explicitly includes economic communities [Colorado General Assembly, 2018].

Arizona Proposition 106 (2000). Arizona's Independent Redistricting Commission criteria list "communities of interest" including economic and social interests [Arizona Secretary of State, 2000].

Washington RCW 44.05.090. Washington's redistricting law requires preservation of "communities of related and mutual interests," with economic community membership recognized in Commission practice [Washington State Legislature, 1983].

At the federal level, *Rucho v. Common Cause* [2019] established that federal courts cannot adjudicate partisan gerrymandering claims. Economic character weights are relevant to this context because they provide a *partisan-neutral* rationale for district lines: the weight modification uses no electoral data, no voter registration data, and no racial or ethnic composition data. The inputs are NAICS-sector job counts from a federal government administrative dataset (LODES WAC) that are published annually by the Census Bureau's LEHD program.

5.2 The Partisan-Neutrality Defense

The legal defensibility of algorithmic redistricting depends in part on demonstrating that the inputs to the algorithm are partisan-neutral. Economic character weights satisfy this requirement structurally:

- **Data source.** LEHD LODES WAC data reports workplace employment counts by NAICS sector. No racial, ethnic, or partisan data is present in the WAC file.
- **Computation.** The economic character vector $\mathbf{e}(t) = (\mathrm{CI}(t), \mathrm{IF}(t), \mathrm{JPR}(t))$ is computed purely from job counts and population. No electoral geography, no voter file, no precinct boundary data enters the computation.
- **Causation direction.** The economic character signal influences district boundaries only through the economic profile of tracts, not through their partisan voting history. Any partisan effect is a consequence of the geographic correlation between economic activity and partisan sorting—a correlation that exists independently of the algorithm's choices.

This structure allows an expert witness to testify that "district boundaries follow natural economic zone transitions, not partisan boundaries," which is legally stronger than asserting a proportionality outcome.

5.3 Expert Witness Framing

For expert witnesses using economic character weights in a redistricting proceeding, we recommend the following framing elements:

1. **Data provenance.** "The edge weight modification uses the Census Bureau's LODES Workplace Area Characteristics data, a federal administrative dataset published annually by the LEHD program. The data records the number of jobs by NAICS sector

in each census block. No electoral, racial, or demographic data enters the weight computation."

2. **Criterion alignment.** "Preserving economic community character serves the [state]'s stated redistricting criterion of maintaining communities of interest. Tracts with similar economic profiles—both residential, both commercial, both industrial—are kept together where possible; district cuts are drawn at economic zone transitions, which are natural community boundaries."

3. **Mechanistic transparency.** "The weight formula assigns a cosine similarity score between 0 and 1 to each pair of adjacent tracts based on their economic character vectors. High similarity (similar economic profile) makes the edge harder to cut; low similarity (different economic profiles) makes the edge easier to cut. The $\alpha = 0.5$ blend factor ensures that geographic contiguity remains the dominant consideration."

4. **State-specificity.** "In North Carolina, this modification reduces the degree to which major employment centers (Research Triangle Park, Charlotte commercial corridor) are embedded within residential partisan districts, producing districts that more cleanly separate employment zones from residential zones. In states without concentrated employment clusters, the modification is appropriately neutral."

5.4 When to Use Economic Character Weights

Economic character weights are most valuable, and most legally supportable, in states that satisfy the following conditions:

- The state has an explicit communities-of-interest criterion in its redistricting law or commission criteria.
- The state has concentrated employment centers (university research parks, hospital campuses, industrial corridors) that are currently being split by district lines drawn primarily on geographic compactness grounds.
- LODES data coverage for the state is complete (all states are covered).
- The practitioner can demonstrate, from the LODES data, that the employment concentration pattern exists and that the proposed plan respects it.

Conversely, in states like Wisconsin where economic geography is diffuse, economic character weights add no value and should not be used as a justification for a specific plan; the null result will undermine rather than support the communities-of-interest argument.

6 Conclusion

We have introduced economic character edge weights for congressional redistricting, operationalized through LODES Workplace Area Characteristics data from the Census Bureau's LEHD program. The method computes a three-dimensional economic character vector (commercial intensity, industrial fraction, jobs per resident) for each census tract, then weights adjacency-graph edges by the cosine similarity between adjacent tracts' character vectors. The blend formula preserves geographic boundary length as the dominant signal while modulating it with economic zone structure.

Empirical results on three states establish the method's core behavioral pattern:

- **NC-14**: 14.3 pp proportionality gap improvement as employment centers emerge as natural cut seams.
- **WI-8**: No effect, as expected in a state with diffuse economic geography.
- **TX-38**: Slight 2.6 pp worsening, consistent with industrial zones being embedded in Republican-leaning areas.

The state-specificity of these results is the paper's central practical contribution. Economic character weights should be applied selectively, in states where the economic and partisan geographies are aligned in a way that allows employment-center boundaries to serve as natural community boundaries. They should not be applied indiscriminately across all states as a universal proportionality booster.

The legal analysis establishes that economic character weights are defensible under communities-of-interest criteria in California, Colorado, Arizona, Washington, and other states with explicit economic community language in their redistricting criteria. The partisan-neutral data inputs (LODES WAC, no electoral data) and the transparent algorithmic mechanism support expert witness testimony framing the weight modification as a community-structure criterion, not a partisan adjustment.

Limitations and future work. All results are single-seed (seed=42). Future work should evaluate multi-seed variance to confirm that the NC improvement is robust and not seed-specific. The within-district economic variance metric (mean standard deviation of `jobs_per_resident` across tracts in each district) was not computed in this evaluation run; the M.1 companion paper will track this as the primary metric for the community character framework. Extension to state legislative districts and to other years of LODES data (2002–2021) would provide additional evidence on the method's stability across electoral cycles.

References

John M Abowd, Bryce E Stephens, Lars Vilhuber, Fredrik Andersson, Kevin L McKinney, Marc Roemer, and Simon Woodcock. The LEHD infrastructure files and the creation of the Quarterly Workforce Indicators. Technical Report CES-WP-09-26, U.S. Census Bureau, Center for Economic Studies, 2009. Describes the LEHD infrastructure, including LODES WAC/RAC/OD files.

Micah Altman and Michael P McDonald. *Modeling the Effect of Communities of Interest on Redistricting*. National Conference of State Legislatures, 2011. Working paper.

Arizona Secretary of State. Arizona proposition 106 (2000): Establishing the independent redistricting commission. `https://azredistricting.org/About/IRC-Criteria.asp`, 2000.

California Secretary of State. California proposition 11 (2008): Voters first act. `https://www.sos.ca.gov/elections/ballot-measures/qualified-ballot-measures/approved-measures/proposition-11-2008`, 2008.

Colorado General Assembly. Colorado constitution, article v, section 44: Independent legislative redistricting commission. `https://leg.colorado.gov/redistricting`, 2018.

Giovanni Della-Libera. B.10: Subdivision-respecting redistricting via county co-membership edge weights. Research track B-algorithm, 2024a.

Giovanni Della-Libera. B.14: VRASection: Minority opportunity bisection via vra-aligned edge weights. Research track B-algorithm, 2024b.

Giovanni Della-Libera. BISECT: Algorithmic congressional redistricting via recursive graph bisection. Software, `https://github.com/gdl/bisect`, 2024c.

Christopher D Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, 2008. Section 6.3: The vector space model and cosine similarity.

National Conference of State Legislatures. Redistricting criteria. `https://www.ncsl.org/redistricting-and-census/redistricting-criteria`, 2021.

Jonathan A Rodden. *Why Cities Lose: The Deep Roots of the Urban-Rural Political Divide*. Basic Books, 2019.

Nicholas O Stephanopoulos. Redistricting and the territorial community. *University of Pennsylvania Law Review*, 160(5):1379–1477, 2012.

U.S. Census Bureau. TIGER/Line shapefiles, 2020. https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html, 2020.

U.S. Supreme Court. *Rucho v. Common Cause*, 588 u.s. 684 (2019), 2019.

Washington State Legislature. Washington RCW 44.05.090: Redistricting commission criteria. https://app.leg.wa.gov/rcw/default.aspx?cite=44.05.090, 1983.